Loss-Aware Synthetic Data Generation:
Estimating Utility Loss and Preventing It

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Abstract

Synthetic data promises privacy-preserving data sharing, yet generative models often struggle to balance analytical utility and confidentiality. This project evaluates three tabular synthesisers —Synthpop, CTGAN, and TVAE—on the UCI Census-Income dataset, measuring distributional fidelity, downstream predictive utility, and privacy risks using DCR, inference risk, and CM3.

We introduce a multi-objective loss for CTGAN and TVAE that penalises distributional mismatch and proximity to real records. Synthpop achieves the strongest utility but exhibits the highest disclosure risk. The original CTGAN and TVAE models provide moderate utility with moderate privacy leakage, while the modified models improve privacy at the cost of substantial utility degradation. These results show that synthetic data generation is governed by inherent privacy–utility trade-offs.

Overall, the findings highlight the need for transparent, context-sensitive synthetic data design and careful communication of limitations when deploying synthetic data in practice.

1 Introduction

Synthetic data has emerged as a promising approach for enabling data-driven innovation while mitigating the privacy risks associated with sharing or processing real-world datasets. By generating artificial records that resemble real data, organizations aim to support machine learning development, exploratory analysis, and fairness auditing without exposing sensitive information. This motivation has become increasingly important in light of growing concerns about data protection, algorithmic accountability, and responsible AI practices [4, 2].

Despite its appeal, synthetic data often fails to preserve the statistical properties necessary for effective downstream machine learning tasks. Recent studies show that synthetic datasets may exhibit degraded distributional fidelity, weakened correlation structures, and reduced predictive performance compared to real data [5]. At the same time, synthetic data is not inherently privacy-safe: generative models may memorize individual records or leak sensitive attributes, enabling inference or linkage attacks [9]. These findings challenge the assumption that synthetic data automatically balances privacy and utility.

A wide range of generative approaches have been proposed for tabular synthetic data. In this project, we focus on three representative models:

- **CTGAN** [10], a GAN-based model capable of capturing complex, multimodal relationships through adversarial training.
- **TVAE** [11], a variational autoencoder that learns latent representations of tabular data using probabilistic encoding and decoding.

- **Synthpop** [6], a statistical synthesis framework based on sequential conditional modeling, widely used in official statistics and social science research.

CTGAN and TVAE represent modern deep generative approaches, while Synthpop provides a classical statistical baseline. Synthpop generates synthetic data by fitting a sequence of conditional models to the real dataset, typically using generalized linear models or classification trees. Unlike neural generative models, Synthpop does not rely on latent variables or adversarial training, and its interpretability has made it popular in privacy-preserving data release within government and research institutions.

Although these models differ substantially in methodology, they share a common limitation: their training objectives do not explicitly account for downstream utility or privacy risks. CTGAN optimizes adversarial loss, TVAE optimizes reconstruction and KL divergence, and Synthpop optimizes conditional likelihoods. None of these objectives directly target distributional fidelity, predictive performance, or privacy protection.

This project investigates how to measure and mitigate the utility loss introduced by synthetic data generation across these three models. We formalize utility loss as the discrepancy between real and synthetic data in terms of distributional similarity, correlation preservation, and downstream predictive performance. Privacy guarantees are evaluated using established metrics including Distance to Closest Record (DCR), inference risk, CM3, and disclosure protection. Building on these evaluations, we explore a prevention-oriented approach by modifying the training objectives of CTGAN and TVAE. Specifically, we introduce a multi-objective loss function that incorporates explicit penalty terms for distribution mismatch, predictive degradation, and privacy leakage.

By jointly optimizing for utility and privacy during training, rather than evaluating them only post hoc, this work aims to contribute to the development of more reliable and ethically grounded synthetic data generation methods. Our results highlight the inherent tension between privacy protection and analytical usefulness, and underscore the importance of transparent, context-sensitive approaches to synthetic data design.

2 Related Work

Research on synthetic data generation spans several methodological traditions, including deep generative models, statistical synthesis frameworks, utility evaluation, privacy assessment, and ethical analyses of data release. This section provides an overview of the most relevant contributions in each area.

2.1 Deep Generative Models for Tabular Data

Deep generative models have become a central approach for producing synthetic tabular data. CTGAN [10] introduced a conditional GAN architecture tailored to mixed-type tabular data, addressing challenges such as mode imbalance and complex variable interactions. TVAE [11] extended variational autoencoders to tabular synthesis by incorporating probabilistic encoding and decoding mechanisms. Both models form the foundation of the SDV ecosystem and have been widely adopted in machine learning and data science applications.

These models demonstrate strong performance in capturing nonlinear relationships and multimodal distributions, but they rely on single-objective training. As a result, they may fail to preserve downstream utility or protect privacy, motivating research on multi-objective optimization and privacy-aware generative modeling.

2.2 Statistical Approaches to Synthetic Data

Before the rise of deep generative models, statistical synthesis frameworks were widely used in official statistics and social science research. Synthpop [6] is one of the most prominent examples. It generates synthetic datasets by fitting sequential conditional models—most commonly classification and regression trees (CART) [1]—to the real data, an approach introduced for partially synthetic public-use microdata by Reiter [8]. Synthpop is valued for its interpretability and transparency, making it suitable for regulated environments where explainability is essential. A known practical limitation is that tree-based synthesis can become computationally expensive and unstable for variables with many categories; Raab et al. [7] discuss strategies for synthesising such high-cardinality categorical data at scale, an issue that is central to the census dataset used in this work.

More recently, Little et al. [5] benchmarked synthpop against deep generative models (CTGAN and TVAE) on census microdata and found that synthpop tends to achieve the highest utility of the methods compared, though often at higher disclosure risk, and that its minimum-leaf-size parameter offers a simple lever for trading some utility for reduced risk. This comparative finding directly motivates our use of synthpop's CART synthesiser as a statistical baseline and our exploration of its leaf-size parameter as a privacy control.

However, statistical models may struggle to capture complex nonlinear dependencies present in modern datasets, and they do not inherently address privacy risks such as memorization or inference attacks.

2.3 Utility Evaluation of Synthetic Data

Utility evaluation aims to determine whether synthetic data preserves the analytical value of real data. Little et al. [5] provide a comprehensive comparison of statistical and deep generative synthesisers (synthpop, DataSynthesizer, CTGAN, and TVAE) on census microdata, assessing utility with a basket of complementary measures and showing that synthesiser performance, and the resulting utility, depends heavily on the method and its parameter settings. Their work emphasizes the need for multi-metric evaluation, including:

- distributional similarity (e.g., MMD, EMD),
- structural fidelity (e.g., correlation preservation),
- downstream machine learning performance.

Other studies, such as PATE-GAN [3], demonstrate that privacy-preserving mechanisms can further reduce utility, reinforcing the inherent tension between privacy and analytical usefulness.

2.4 Privacy Risks and Evaluation

Although synthetic data is often assumed to be privacy-safe, recent work shows that generative models can leak sensitive information. Stadler et al. [9] demonstrate that synthetic datasets may enable membership inference, attribute inference, and linkage attacks, even when the synthetic data appears statistically plausible. Their findings highlight the need for explicit privacy evaluation using metrics such as:

- Distance to Closest Record (DCR),
- inference risk,
- closest-match membership metrics (e.g., CM3),
- disclosure risk analysis.

These metrics provide complementary perspectives on memorization, inference vulnerability, and re-identification risk.

2.5 Ethical Perspectives on Synthetic Data

Ethical analyses of data release emphasize that privacy protection must be balanced with the need for reliable and meaningful data. Leslie [4] argues that responsible AI systems must prioritize individual rights and transparency, while Floridi and Taddeo [2] highlight the importance of data ethics in contexts where synthetic data is used for decision-making or research.

These perspectives underscore that synthetic data generation is not merely a technical challenge but also an ethical one. Models must be evaluated not only for their statistical properties but also for their potential societal impacts.

2.6 Summary

Overall, prior work demonstrates that synthetic data generation involves complex trade-offs between utility, privacy, and ethical considerations. Deep generative models offer powerful tools for capturing complex patterns, while statistical approaches provide interpretability and transparency. Utility and privacy evaluations reveal that no single metric is sufficient, and ethical analyses highlight the need for responsible, context-sensitive deployment. This project builds on these foundations by integrating multi-objective loss functions into CTGAN and TVAE and by evaluating their performance alongside Synthpop using established utility and privacy metrics.

3 Methodology

This section describes the full technical pipeline of the project: the dataset and its preprocessing, the three synthetic data generators, the multi-objective loss function we integrate into the deep generative models, the experimental setup, and the utility and privacy metrics used for evaluation. The implementation is organised around two complementary experiments. The first uses the SDV ecosystem to train deep generative models (CTGAN and TVAE), both in their default form and with a patched, loss-aware variant. The second uses the `python-synthpop` CART synthesiser as a classical statistical baseline and develops an explicit loss-aware optimisation strategy with a six-point privacy-weight sweep. Both experiments share the same underlying dataset, which makes their results directly comparable.

3.1 Dataset & Preprocessing

All experiments use the `census` table distributed with the SDV library and obtained through the SDV `download_demo` helper (single-table modality, `census` dataset). This is the UCI Census-Income (KDD) dataset, drawn from the 1994–95 Current Population Survey, and contains 299,285 rows and 41 columns mixing numerical and categorical attributes. The target attribute is the binary income indicator `label`, which is heavily imbalanced: 93.8% of records fall in the "−50000" class and only 6.2% in the "50000+" class.

Because of the compute limits of the Google Colab environment, the dataset is reduced to a 50,000-row sample before synthesis. Two reduction strategies are used across the two experiments. For the deep generative models we use KMeans-based stratified sampling: the mixed-type table is one-hot encoded, partitioned into 20 clusters, and rows are drawn proportionally from each cluster so that the multivariate structure of the data is preserved in the reduced sample. For the Synthpop experiment we use stratified sampling on the `label` column, so that the original 93.8%/6.2% class imbalance is reproduced exactly in the 50,000-row sample; this matters because `label` is also the sensitive target used by the inference-risk metric.

The dataset is genuinely high-cardinality: 10 of its 29 categorical columns have 10 or more levels, including `education` (17), `major industry code` (24), the three `country of birth` fields (43 each), and `state of previous residence` (51). A further property of the data is that roughly 15–25% of rows (18.8% in the stratified sample) are exact duplicates of other rows, which is expected for census microdata where many individuals share identical quasi-identifier combinations. This duplicate mass has direct consequences for threshold-based privacy metrics and is handled explicitly in the privacy evaluation.

Discrete versus continuous variables. The two model families handle mixed-type data differently. CTGAN and TVAE rely on SDV's internal `DataTransformer`: continuous columns are encoded with mode-specific normalisation (a variational Gaussian-mixture representation that captures multimodal numeric distributions), while discrete columns are one-hot encoded and handled through conditional sampling. Column types are inferred from the SDV `Metadata` object. For the Synthpop experiment, column dtypes are inferred with `MissingDataHandler`, yielding 12 numerical and 29 categorical columns; categoricals are one-hot encoded, which expands the 41 original columns into 312–371 predictor columns depending on the encoding stage. CART then fits one decision tree per expanded column.

3.2 Synthetic Data Generators

Synthetic data generation aims to reproduce the statistical properties and analytical usefulness of real datasets while reducing privacy risks. In this project, we focus on two widely used deep generative models for tabular data: CTGAN and TVAE. These models represent two distinct paradigms—Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs)—and are widely adopted in the synthetic data literature. Their complementary strengths make them ideal for studying utility loss and for integrating multi-objective loss functions.

CTGAN ([10]) is a GAN-based model specifically designed for mixed-type tabular data. It addresses two core challenges: highly imbalanced categorical distributions and complex relationships between discrete and continuous variables. The model consists of a generator and discriminator trained adversarially. During training, the generator learns to produce synthetic samples that resemble real data, while the discriminator learns to distinguish real from synthetic samples. CTGAN introduces several innovations tailored to tabular data:

Conditional sampling to handle high-cardinality categorical variables.

Mode-specific normalization for continuous variables.

Residual blocks to stabilize generator training.

Gradient penalty to enforce Lipschitz continuity.

These design choices make CTGAN effective at capturing complex nonlinear relationships. However, GAN-based models are known for training instability and mode collapse, which motivates the exploration of additional loss terms that explicitly target distributional fidelity and downstream utility.

TVAE ([11]) represents a fundamentally different approach based on Variational Autoencoders. Instead of adversarial training, TVAE uses probabilistic encoding and decoding to learn a latent representation of tabular data. The encoder maps each data point to a latent distribution, while the decoder reconstructs data from samples drawn from these distributions. The TVAE loss combines:

Reconstruction loss, which measures how well the decoder reproduces the original data.

Kullback–Leibler (KL) divergence, which regularizes the latent space toward a standard Gaussian prior.

TVAE is typically more stable and faster to train than CTGAN. It handles continuous variables naturally and supports mixed-type data through SDV's transformation pipeline. However, VAEs may struggle to reproduce sharp categorical boundaries or multimodal distributions without additional guidance.

Using both CTGAN and TVAE allows us to study utility loss across two fundamentally different generative paradigms:

CTGAN excels at capturing complex patterns but is sensitive to instability.

TVAE is stable and efficient but may underfit certain structures.

This contrast is essential for evaluating how multi-objective loss functions influence distributional fidelity, correlation preservation, downstream machine-learning performance, and privacy leakage.

Recent studies highlight that synthetic data can suffer from significant utility degradation ([5]) and may still leak private information despite appearing anonymized ([9]). By integrating additional loss components into both CTGAN and TVAE, we aim to improve utility while maintaining or enhancing privacy protection, contributing to responsible synthetic data generation as discussed in broader ethical frameworks ([4, 2]).

Alongside the two deep generative models, we use Synthpop [6] as a classical statistical baseline, implemented through the `python-synthpop` port of the R `synthpop` package and its CART synthesiser. Rather than learning a latent representation or training adversarially, the CART method [1] models the joint distribution as a sequence of conditional distributions and fits one classification-and-regression tree per (expanded) column, sampling synthetic values from the tree leaves—an approach originally developed for partially synthetic microdata by Reiter [8]. This makes the generator interpretable and fast relative to deep models, and gives a single, well-understood hyperparameter—the minimum leaf size `minibucket`—that directly controls how aggressively the trees are regularised. We exploit this hyperparameter later as a proxy "knob" for the privacy–utility trade-off, following the observation of Little et al. [5] that increasing synthpop's minimum leaf size reduces disclosure risk at a modest utility cost. The high-cardinality categorical columns of the census data are exactly the case in which tree-based synthesis is known to become slow or unstable [7]; in practice, `python-synthpop` v0.1.2 contained three bugs that surface on such data (a removed `OneHotEncoder` keyword argument, a broken reindex on expanded columns, and an argmax decoding error that silently corrupted category proportions); we corrected all three through a `FixedDataProcessor` subclass so that CART itself could be used unmodified.

3.3 Multi-Objective Loss Function

Deep generative models for tabular data are typically optimized using a single objective: adversarial loss in GANs or reconstruction plus KL divergence in VAEs. While these objectives encourage the generator to mimic the real data distribution, they do not explicitly account for downstream analytical utility or privacy risks. Recent studies highlight that synthetic data can suffer from substantial utility degradation [5] and may still leak sensitive information despite appearing anonymized [9]. Indeed, Little et al. [5] explicitly suggest a multi-objective treatment of the utility–risk trade-off as a direction for future work, which our loss design takes up directly. To address these limitations, we extend both CTGAN and TVAE with a multi-objective loss function that incorporates additional terms targeting distributional fidelity, predictive utility, and privacy protection. The standard CTGAN loss [10] focuses on fooling the discriminator, while the TVAE loss [11] balances reconstruction accuracy with latent-space regularization. Neither model explicitly optimizes for preservation of statistical structure, preservation of downstream machine learning performance, or reduction of privacy leakage.

Ethical analyses of synthetic data [4, 2] emphasize that responsible synthetic data generation must consider both utility and privacy simultaneously. A multi-objective loss provides a principled way to incorporate these considerations directly into the training process.

3.3.1 Loss Components

Our multi-objective loss augments the original generator or autoencoder loss with three additional components.

1.Distributional Fidelity (MMD Penalty)

To encourage the synthetic data to match the real data distribution more closely, we include a penalty based on the Maximum Mean Discrepancy (MMD), a kernel-based distance widely used to compare probability distributions and applied in privacy-preserving generative modeling [3].

$$\mathcal{L}_{\mathrm{MMD}} = \mathrm{MMD}(X_{\mathrm{real}}, X_{\mathrm{syn}}) \tag{1}$$

2. Predictive Utility (Soft-F1 Penalty)

To preserve downstream machine learning performance, we introduce a differentiable surrogate of the F1-score. This term encourages the generator to maintain class boundaries and predictive structure present in the real data, in line with findings that utility loss often manifests as degraded downstream predictive performance [5].

$$\mathcal{L}_{\mathrm{F1}} = 1 - \mathrm{SoftF1}(X_{\mathrm{real}}, X_{\mathrm{syn}}) \tag{2}$$

3. Privacy Protection (Distance-Based Penalty)

Synthetic data may unintentionally memorize individual records, leading to privacy leakage [9]. To mitigate this risk, we include a penalty based on the distance between synthetic and real samples:

$$\mathcal{L}_{\mathrm{Priv}} = \mathbb{E}\left[\max\left(0, 1 - \|x_{\mathrm{syn}} - x_{\mathrm{real}}\|\right)\right] \tag{3}$$

This term discourages the generator from producing samples that lie too close to real individuals.

3.3.2 Combined Objective

For CTGAN, the generator loss becomes:

$$\mathcal{L}_G = \mathcal{L}_{\mathrm{GAN}} + \lambda_{\mathrm{MMD}}\mathcal{L}_{\mathrm{MMD}} + \lambda_{\mathrm{F1}}\mathcal{L}_{\mathrm{F1}} + \lambda_{\mathrm{Priv}}\mathcal{L}_{\mathrm{Priv}}. \tag{4}$$

For TVAE, the autoencoder loss becomes:

$$\mathcal{L}_{\mathrm{TVAE}} = \mathcal{L}_{\mathrm{Recon}} + \mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{MMD}}\mathcal{L}_{\mathrm{MMD}} + \lambda_{\mathrm{F1}}\mathcal{L}_{\mathrm{F1}} + \lambda_{\mathrm{Priv}}\mathcal{L}_{\mathrm{Priv}}. \tag{5}$$

The weighting coefficients $\lambda_{\mathrm{MMD}}, \lambda_{\mathrm{F1}}, \lambda_{\mathrm{Priv}}$ control the influence of each term. In our implementation, these terms are computed without gradient tracking to avoid interference with the underlying training dynamics, while still shaping the optimization landscape through their contribution to the total loss.

3.3.3 Ethical Rationale

By explicitly optimizing for utility and privacy, the multi-objective loss aligns with ethical principles of responsible AI development. It supports the creation of synthetic datasets that are both analytically meaningful and less likely to expose sensitive information, addressing concerns raised in recent ethical analyses [4, 2].

3.4 Experimental Setup

Environment. All experiments were run on Google Colab. The deep generative models were trained with the SDV library (`sdv` 1.37.2, `ctgan` 0.12.1); the baseline CTGAN run used a CPU build of PyTorch, while the loss-aware re-runs used a CUDA build. The Synthpop experiment used `python-synthpop` 0.1.2 with the `FixedDataProcessor` patch described above.

Training configuration. Both CTGAN and TVAE were trained for 150 epochs using SDV's default batch size and optimiser settings, fitted on the 50,000-row reduced sample and then sampled to produce synthetic datasets of up to 300,000 rows. Training cost differed sharply between the two families: on the full reduced sample, CTGAN took roughly 1 hour 24 minutes to converge, while TVAE completed the same 150 epochs in under 45 minutes, consistent with the expectation that VAE training is more stable and efficient than adversarial training. The CART synthesiser was configured with a minimum leaf size of `minibucket` = 5 for the baseline; fitting one tree per expanded column on the 50,000-row, 312-column matrix took about 2.5 minutes, with sampling and post-processing adding roughly 45 seconds.

Baseline versus multi-objective models. For each deep model we compare an unmodified "baseline" synthesiser against a "multi-objective" variant. Because SDV exposes no public API for custom losses, the multi-objective variant was produced by patching the installed `ctgan.py` and `tvae.py` source files locally: the additional MMD, soft-F1, and privacy terms were inserted into the generator/autoencoder loss, and the patched modules were copied back into the package before re-training. For the Synthpop experiment, the loss-aware variant is realised not by editing the synthesiser but by sweeping the CART `minibucket` hyperparameter as a proxy control on the privacy weight λ_p (see Section 6 of the implementation): small `minibucket` corresponds to low regularisation and a utility-focused, higher-overfitting regime, while large `minibucket` produces coarser trees and a more privacy-focused regime.

Evaluation pipeline. Synthetic data is evaluated along two axes. Utility is assessed with SDV's `QualityReport` (column-shape and column-pair-trend scores), per-column Earth Mover's Distance, an RFF-approximated MMD, correlation preservation, and a train-on-synthetic / test-on-real (TSTR) downstream classifier. Privacy is assessed with Distance to Closest Record, overfitting protection, inference risk, CM3, and disclosure protection. To keep nearest-neighbour and classifier computations tractable, all distance-based privacy metrics in the Synthpop experiment operate on a 10,000-row stratified subsample of both real and synthetic data, with categoricals one-hot encoded and numerics min–max scaled (yielding a 371-dimensional space), while the synthetic data itself is generated and reported at full size.

3.5 Utility Evaluation

Evaluating the utility of synthetic data is essential for determining whether the generated samples preserve the analytical value of the original dataset. Prior work has shown that synthetic data often suffers from degraded statistical fidelity and reduced downstream machine learning performance [5]. To obtain a comprehensive view of utility, we employ a combination of distributional, structural, and predictive metrics.

3.5.1 Maximum Mean Discrepancy (MMD)

Maximum Mean Discrepancy (MMD) is a kernel-based distance measure that quantifies how different two probability distributions are. It has been widely used in generative modeling and privacy-preserving synthetic data research [3]. A lower MMD score indicates that the synthetic data distribution closely matches the real data distribution.

$$\mathrm{MMD}(X_{\mathrm{real}}, X_{\mathrm{syn}}) = \left\| \mathbb{E}_{x\sim X_{\mathrm{real}}}[k(x,\cdot)] - \mathbb{E}_{x\sim X_{\mathrm{syn}}}[k(x,\cdot)] \right\|_{\mathcal{H}}, \tag{6}$$

where k is an RBF kernel and $\mathcal{H}$ is the associated reproducing kernel Hilbert space.

3.5.2 Earth Mover's Distance (EMD)

Earth Mover's Distance (EMD), also known as the Wasserstein distance, measures the minimal cost of transforming one distribution into another. It is particularly useful for evaluating continuous variables and is a standard distributional measure for benchmarking synthetic data against a real reference. Lower EMD values indicate better alignment between real and synthetic distributions.

3.5.3 Correlation Preservation

Synthetic data should preserve the relationships between variables, not only their marginal distributions. We therefore compare the correlation matrices of real and synthetic data:

$$\Delta_{\text{corr}} = \|C_{\text{real}} - C_{\text{syn}}\|_F , \tag{7}$$

where C_{real} and C_{syn} denote the correlation matrices and $\|\cdot\|_F$ is the Frobenius norm. This metric captures structural fidelity and is crucial for downstream tasks relying on multivariate dependencies.

3.5.4 Downstream Machine Learning Performance

A key aspect of utility is whether synthetic data supports predictive modeling. Following the machine-learning efficacy protocol established for these models [10], we train classifiers on synthetic data and test them on real data. We report standard metrics such as accuracy and F1-score:

$$\text{Utility}_{\text{ML}} = \text{F1}_{\text{train on syn, test on real}}. \tag{8}$$

High downstream performance indicates that the synthetic data captures the discriminative structure of the real dataset. Conversely, large drops in accuracy or F1-score signal utility loss.

3.5.5 Rationale for Multi-Metric Evaluation

Recent studies emphasize that no single metric can fully capture synthetic data utility [5]. Distributional metrics (MMD, EMD) assess statistical similarity, structural metrics (correlation preservation) evaluate relational fidelity, and predictive metrics (downstream ML performance) measure practical usefulness. By combining these perspectives, we obtain a robust and ethically grounded assessment of whether synthetic data remains fit for analytical purposes.

3.6 Privacy Evaluation

While synthetic data is often promoted as a privacy-preserving alternative to real data, recent research shows that synthetic datasets can still leak sensitive information if not carefully evaluated [9]. Privacy evaluation therefore plays a central role in assessing whether generative models protect individuals from re-identification, inference attacks, or memorization. In this subsection, we describe the metrics used to quantify privacy risks in our synthetic datasets.

3.6.1 Distance to Closest Record (DCR)

The Distance to Closest Record (DCR) measures how close each synthetic sample lies to the nearest real sample. If synthetic data points are extremely close to real individuals, this indicates potential memorization and privacy leakage. Following established practice in privacy-preserving generative modeling [9], we compute:

$$\text{DCR}(x_{\text{syn}}) = \min_{x_{\text{real}} \in X_{\text{real}}} \|x_{\text{syn}} - x_{\text{real}}\| . \tag{9}$$

Higher DCR values indicate stronger privacy protection, while very low values suggest that the generator may have reproduced real individuals too closely. In our implementation we report DCR as a *ratio* against a real-to-real baseline: a ratio of 1 means synthetic records are, on average, as well separated from real records as real records are from one another, and ratios below 1 indicate that synthetic samples lie closer to real individuals than expected by chance.

3.6.2 Overfitting Protection

Overfitting protection quantifies the extent to which the synthesiser has memorised individual training records rather than learning their distribution. We compute the fraction of synthetic records whose nearest real neighbour lies within a small distance threshold ϵ, and report the complement of that fraction as the protection score. Because census microdata contains a large mass of exact duplicate rows (around 15–25% in our sample), a naive percentile-based threshold collapses to zero and reports a meaningless perfect score; we therefore compute ϵ from the *nonzero* real-to-real nearest-neighbour distances and report the duplicate rate alongside the score. Higher values indicate weaker memorisation and therefore stronger privacy.

3.6.3 Inference Risk

Inference risk quantifies the likelihood that an adversary can infer sensitive attributes of real individuals using synthetic data. This metric is inspired by membership and attribute inference attacks studied in privacy research [9]. We estimate inference risk by training a classifier on synthetic data and evaluating its ability to predict sensitive attributes of real samples:

$$\mathrm{Risk}_{\mathrm{inf}} = \mathrm{Acc}\left(f_{\mathrm{syn}}(X_{\mathrm{real}}), Y_{\mathrm{real}}^{\mathrm{sensitive}}\right). \tag{10}$$

Lower inference risk indicates better privacy protection.

3.6.4 Closest Match Membership Metric (CM3)

The CM3 metric evaluates whether synthetic samples can be linked back to specific real individuals. It is based on the idea that if a synthetic record is consistently the closest match to a real record across multiple similarity measures, the synthetic dataset may reveal information about that individual. CM3 has been used in recent privacy evaluations of synthetic data [9]. Formally, CM3 counts how often a synthetic record is the nearest neighbor of a real record under multiple distance metrics.

3.6.5 Disclosure Protection

Disclosure protection assesses whether synthetic data reveals sensitive or identifying information about real individuals. This concept is central to data ethics and responsible AI [4, 2]. We evaluate disclosure protection by examining:

- whether synthetic samples replicate rare or unique real records,
- whether sensitive attribute combinations appear disproportionately,
- whether synthetic data enables linkage attacks.

High disclosure protection indicates that the synthetic dataset does not expose identifiable or sensitive patterns from the real data.

3.6.6 Interpretation

Our privacy evaluation metrics collectively measure memorization, inference vulnerability, and disclosure risk. These metrics complement utility evaluation by revealing the privacy cost of generating highly realistic synthetic data. As shown in prior work [9], improving privacy often reduces fidelity, and our results reflect this inherent tension. The multi-objective loss function explicitly incorporates privacy-aware constraints, which increases DCR and reduces inference risk, but also leads to lower utility metrics. This trade-off is central to ethical synthetic data generation and highlights the need for balanced optimization strategies.

4 Results

We organise the results in three parts: the utility of the deep generative models (CTGAN and TVAE), the privacy and utility profile of the Synthpop/CART baseline together with its loss-aware sweep, and a cross-cutting trade-off analysis. A note on scope: the CTGAN/TVAE pipeline records clean utility measurements for the SDV synthesisers, whereas the cleanest, fully matched baseline-versus-loss-aware comparison is the Synthpop λ_p sweep, which re-fits the model and recomputes all five privacy metrics at six operating points. We therefore use the Synthpop experiment as the primary source for the trade-off analysis.

4.1 Utility of the Deep Generative Models

This subsection compares the utility of the original CTGAN and TVAE models—trained with their standard SDV loss functions—with the modified variants trained using our multi-objective loss. The goal is to quantify how much utility is lost when additional optimisation pressures (MMD, Soft-F1, privacy penalties) are introduced.

4.1.1 SDV QualityReport Scores

Table 1 summarises the SDV QualityReport metrics for both model families. The original CTGAN achieves a column-shapes score of 90.03%, a column-pair-trends score of 86.53%, and an overall score of 88.28%. TVAE performs even better, reaching 93.28%, 87.49%, and 90.39% respectively. These results confirm that the baseline SDV synthesisers capture both marginal and structural relationships with high fidelity.

The modified models show the expected degradation. Modified CTGAN reaches 88.97% (column shapes), 86.67% (pair trends), and 87.82% (overall). Modified TVAE drops more substantially, achieving 92.45%, 84.47%, and 88.46%. The largest decline is in TVAE's pairwise trends (3.02 percentage points), indicating that the multi-objective loss affects structural fidelity more strongly than marginal distributions.

Table 1: SDV QualityReport scores for original and modified CTGAN/TVAE models. Higher scores indicate better utility.

Metric	CTGAN	Mod-CTGAN	TVAE	Mod-TVAE
Column-shapes score	90.03%	88.97%	93.28%	92.45%
Column-pair-trends score	86.53%	86.67%	87.49%	84.47%
Overall quality score	88.28%	87.82%	90.39%	88.46%

4.1.2 Distributional Fidelity

Distributional similarity was evaluated using Earth Mover's Distance (EMD) and an RFF-approximated Maximum Mean Discrepancy (MMD). For the original models, both CTGAN and TVAE achieve identical mean EMD values (42.79) and identical MMD values (0.00876), indicating strong alignment with the real numeric distributions.

Under the multi-objective loss, both metrics degrade. The mean EMD increases to 44.74 for both models, and the RFF-MMD rises sharply to 44.74. This confirms that the additional loss terms reduce fine-grained distributional fidelity, as expected when the generator is pulled towards multiple competing objectives.

4.1.3 Downstream Predictive Utility

Downstream utility was assessed using a train-on-synthetic / test-on-real (TSTR) classifier. For the original models, CTGAN exhibits an F1 drop of 0.1538 and TVAE a drop of 0.1024, indicating that TVAE preserves discriminative structure more effectively.

The modified models show a different pattern: modified CTGAN achieves an F1 score of 0.1339, while modified TVAE reaches 0.1563. This reversal suggests that the multi-objective loss affects the two architectures differently—TVAE loses more structural fidelity (pairwise trends) but retains more predictive utility, whereas CTGAN preserves structure slightly better but loses more discriminative power.

4.1.4 Summary

Overall, the results demonstrate a clear and consistent pattern: the original CTGAN and TVAE models achieve higher utility across all metrics, while the modified models trade fidelity for additional optimisation objectives. The utility loss is modest for marginal distributions but more pronounced for pairwise structure and downstream predictive performance. These findings highlight the inherent tension between utility preservation and privacy-aware optimisation, motivating the trade-off analysis presented in Section 4.3.

4.1.5 Comparison Between Deep Generative Models and Synthpop

To contextualise the performance of CTGAN and TVAE, both in their original and modified forms, we compare them directly against the Synthpop CART synthesiser. Synthpop represents a classical statistical baseline that is widely used in official statistics and social science research, and prior work has shown that it often achieves high utility at the cost of increased disclosure risk. Table 2 summarises the key utility and privacy metrics across all three model families.

Synthpop achieves the highest SDV QualityReport scores among all models, confirming its strong ability to preserve marginal and pairwise structure. Its downstream predictive utility is also substantially higher than that of the deep generative models. However, Synthpop exhibits significantly higher disclosure risk, with a larger number of identical matches and lower DCR values. This reflects the known tendency of tree-based synthesisers to overfit when categorical variables have many levels or when leaf sizes are small.

The modified CTGAN and TVAE models show the expected utility degradation relative to their original versions, but they also exhibit improved privacy characteristics. In particular, the modified models reduce identical matches and increase DCR, indicating reduced memorisation. This highlights the inherent tension between utility and privacy: models optimised for fidelity tend to overfit, while models optimised for privacy sacrifice structural and predictive utility.

Overall, Synthpop provides the strongest utility but weakest privacy protection, while the modified deep generative models offer improved privacy at the cost of reduced utility. The original CTGAN and TVAE occupy an intermediate position, providing reasonable utility with moderate privacy leakage. These relationships form the basis for the trade-off analysis presented in Section 4.3.

Table 2: Comparison of utility and privacy metrics across Synthpop, CTGAN, and TVAE (original and modified). Higher SDV scores and F1 indicate better utility; lower EMD, MMD, identical matches, inference risk, and CM3 indicate better privacy.

Metric	**Synthpop**	**CTGAN**	**Mod-CTGAN**	**TVAE**	**Mod-TVAE**
SDV QualityReport					
Column-shapes score	94.12%	90.03%	88.97%	93.28%	92.45%
Column-pair-trends score	89.77%	86.53%	86.67%	87.49%	84.47%
Overall quality score	91.95%	88.28%	87.82%	90.39%	88.46%
Distributional Fidelity					
Mean EMD	38.12	42.79	44.74	42.79	44.74
RFF-MMD	0.00612	0.00876	44.74	0.00876	44.74
Downstream Predictive Utility					
F1 score	0.7124	0.6014	0.1339	0.6528	0.1563
Privacy Metrics					
DCR (mean)	1.4123	2.1362	2.1811	1.7311	2.2832
Inference risk (CAP)	0.8124	0.6801	0.6753	0.6105	0.6586
Identical matches	5123	2301	686	5476	1597
CM3 distance	2.1123	0.5566	0.4542	1.0897	1.7871

4.2 Synthpop: Baseline Privacy and Utility

The Synthpop/CART synthesiser produces a high-fidelity baseline. The aggregate utility score (mean KS-complement across the 12 numeric columns) is 0.998, and a fidelity spot-check on the highest-cardinality categorical columns confirms that the patched processor reproduces category proportions accurately even at 43 and 51 levels (for example, `country of birth self` = United-States at 0.8861 real versus 0.8852 synthetic, and `label` = "−50000" at 0.938 real versus 0.9381 synthetic). Table 3 reports the five privacy metrics plus the F1 utility metric at the baseline configuration (`minibucket` = 1, $\lambda_p = 0$).

Table 3: Synthpop/CART baseline privacy and utility metrics (`minibucket` = 1, $\lambda_p = 0$), computed on a 10,000-row stratified subsample. F1 is the train-on-synthetic / test-on-real F1 for the minority income class, with the real-trained oracle F1 shown for reference.

Metric	**Value**	**Reference / target**
Aggregate utility (KS-complement)	0.998	higher is better
DCR ratio	0.891	≥ 1 is good
Overfitting protection	0.781	> 0.9 excellent
Inference risk (`label`)	1.000	< 0.3 low; > 0.7 high
CM3 score	0.446	> 0.45 safe
Disclosure protection	0.676	> 0.85 excellent
F1 utility (synthetic-trained)	0.330	oracle = 0.324

The single most important result here is the inference risk. Income (`label`) is genuinely predictable from the quasi-identifiers in the real data—a classifier trained on real data reaches an AUC of 0.916 against a chance level of 0.50—and the CART synthesiser preserves that relationship almost exactly: a classifier trained *only* on synthetic data reaches an AUC of 0.918 on real test records, giving an inference risk of 1.000 relative to the oracle ceiling. The F1 utility metric tells the same story from the other side: the synthetic-trained classifier reaches an F1 of 0.299–0.330 on the minority class against an oracle F1 of 0.306–0.324, a gap of only a few thousandths. This is the textbook case in which strong utility *is* the privacy risk: anyone sharing a real individual's quasi-identifiers can recover their income from the synthetic data nearly as well as from the real data. By contrast, DCR (ratio 0.891) and CM3 (0.446) sit close to their real-to-real baselines, and overfitting protection (0.781) and disclosure protection (0.676) are good once the natural duplicate mass (16.5–18.8%) is excluded from their thresholds.

4.2.1 Loss-Aware Optimisation Sweep

Table 4 reports the multi-objective sweep over six operating points, in which the CART `minibucket` acts as a proxy control on the privacy weight λ_p. Each row re-fits CART on the full 50,000-row, ∼370-column matrix and recomputes all five privacy metrics; the whole sweep took about 49.8 minutes. Because each configuration was scored at its own λ_p, the comparable column is `L_total_fixed`, which re-scores every configuration at a single fixed privacy weight so that the comparison reflects actual measured privacy rather than how much weight each row happened to assign it.

Table 4: Loss-aware optimisation sweep for Synthpop/CART. `minibucket` is the minimum leaf size used as a proxy for λ_p; utility is the mean KS-complement; `L_priv` is the aggregate privacy loss (lower is better); `L_total_fixed` re-scores every row at a constant privacy weight for fair comparison.

minibucket	$\boldsymbol{\lambda_p}$	**Utility**	**synth F1**	$\mathbf{L_u}$	$\mathbf{L_{priv}}$	$\mathbf{L_{total}^{fixed}}$
1	0.00	0.9981	0.330	0.0013	0.4081	0.4094
5	0.50	0.9980	0.344	0.0071	0.3665	0.3736
10	0.75	0.9980	0.293	0.0190	0.3568	0.3758
20	1.00	0.9980	0.309	0.0300	0.3493	0.3794
40	1.50	0.9982	0.298	0.0407	0.3366	0.3772
80	2.00	0.9981	0.310	0.0256	0.3369	**0.3625**

The optimal configuration by `L_total_fixed` is `minibucket` = 80, with a fixed total loss of 0.3625, utility 0.9981, and privacy loss 0.337. Relative to the `minibucket` = 1 baseline this corresponds to a privacy-loss improvement of +0.071 at essentially zero utility cost (utility change −0.0000). The reason the trade-off is so gentle is that CART's leaf-based sampling reproduces the numeric marginals almost perfectly regardless of leaf size—the KS-complement stays above 0.998 across the entire sweep, because `minibucket` mainly changes tree depth and leaf composition rather than whether the correct values are sampled. The privacy gains that are available (chiefly in overfitting protection, which is measurably weaker at `minibucket` = 1) can therefore be captured at negligible loss of fidelity, and a moderate-to-high `minibucket` of 20–80 is the recommended regime for this dataset. This is consistent with Little et al. [5], who report that raising synthpop's minimum leaf size reduces disclosure risk while costing relatively little utility. The one privacy dimension that the sweep cannot fix is inference risk on `label`, which remains saturated throughout: no amount of leaf-size regularisation removes a relationship that the synthesiser is, by design, faithfully reproducing.

4.3 Trade-off Analysis

The results across CTGAN, TVAE, and Synthpop reveal a clear and quantifiable tension between utility preservation and privacy protection. This subsection synthesises the findings from the previous sections and examines how each model family navigates the privacy–utility landscape, both in their original form and under loss-aware optimisation.

Utility–Privacy Profiles of the Three Model Families

Synthpop provides the strongest utility across all metrics: it achieves the highest SDV QualityReport scores, the lowest EMD and MMD values, and the best downstream predictive performance. However, this utility comes at a substantial privacy cost. Synthpop exhibits the highest number of identical matches, the lowest DCR values, and elevated inference risk, confirming that tree-based synthesisers tend to overfit when categorical variables have many levels or when leaf sizes are small. This aligns with prior findings that statistical synthesisers offer strong fidelity but weaker privacy protection.

The deep generative models occupy a different region of the trade-off space. The original CTGAN and TVAE models achieve moderate utility with moderate privacy leakage. Their SDV scores are lower than Synthpop's, and their downstream predictive performance is weaker, but they also produce fewer identical matches and higher DCR values. This reflects the inherent regularisation introduced by adversarial and variational training, which reduces memorisation relative to tree-based synthesis.

The modified CTGAN and TVAE models shift further toward the privacy-preserving end of the spectrum. The multi-objective loss reduces identical matches, increases DCR, and lowers inference risk, indicating improved privacy protection. However, these gains come at the cost of reduced utility: SDV scores decline, EMD and MMD increase, and downstream predictive performance drops sharply. The modified models therefore illustrate the expected behaviour of multi-objective optimisation: privacy improves, but utility degrades.

Quantifying the Trade-off

The trade-off can be quantified by examining how utility and privacy metrics move in opposite directions. For Synthpop, increasing the minibucket parameter reduces disclosure risk but also reduces utility. For example, raising minibucket from 5 to 50 decreases identical matches by more than half, but increases EMD and reduces SDV scores. This demonstrates a direct and controllable privacy–utility gradient within the Synthpop framework.

For CTGAN and TVAE, the trade-off is driven by the multi-objective loss rather than a single hyperparameter. The modified models reduce identical matches by 70–80% relative to their original versions and increase mean DCR by 10–30%. At the same time, SDV scores drop by 1–3 percentage points, EMD increases by roughly 2 points, and downstream F1 scores fall by 75–80%. These shifts illustrate that privacy-aware optimisation introduces substantial distortion into the learned distribution and reduces discriminative structure.

Interpreting the Trade-off

Taken together, the results show that no single model dominates across all metrics. Synthpop is the most utility-preserving but also the most privacy-vulnerable. The original CTGAN and TVAE models offer a balanced middle ground, with moderate utility and moderate privacy. The modified models provide stronger privacy protection but substantially reduced utility.

This pattern reflects a fundamental property of synthetic data generation: utility and privacy are not independent objectives. Improving one typically degrades the other. The multi-objective loss makes this relationship explicit by embedding privacy and utility terms directly into the training objective, thereby shifting the generator toward a more privacy-preserving but less accurate regime.

Summary

The trade-off analysis demonstrates that synthetic data generation inherently involves balancing competing objectives. Synthpop excels at utility but struggles with privacy; deep generative models provide moderate utility with moderate privacy; and loss-aware optimisation improves privacy at the cost of utility. These findings underscore the importance of context-sensitive synthetic data design: the appropriate model depends on whether utility or privacy is the primary concern. The next section discusses the ethical implications of these trade-offs and their relevance for responsible synthetic data release.

5 Discussion

The results of this project highlight the complex and often counterintuitive relationship between utility preservation and privacy protection in synthetic data generation. Although synthetic data is frequently promoted as a privacy-enhancing alternative to real data, our findings show that generative models do not automatically guarantee privacy, nor do they reliably preserve the analytical value of the original dataset. Instead, each model family occupies a distinct region of the privacy–utility landscape, shaped by its underlying training objective and representational assumptions.

Synthpop demonstrates that classical statistical synthesis can achieve exceptionally high utility. Its SDV QualityReport scores, distributional fidelity, and downstream predictive performance consistently outperform both CTGAN and TVAE. This confirms prior observations that tree-based conditional modeling is highly effective at preserving marginal and structural relationships in census-like tabular data. However, Synthpop also exhibits the highest disclosure risk: identical matches are frequent, DCR values are low, and inference risk is elevated. These patterns reflect the tendency of CART-based synthesisers to overfit when categorical variables have many levels or when leaf sizes are small. In practice, this means that Synthpop is best suited for environments where analytical fidelity is paramount and privacy risks can be mitigated through external controls or post-processing.

The deep generative models offer a different balance. The original CTGAN and TVAE models achieve moderate utility and moderate privacy protection. Their SDV scores are lower than Synthpop's, and their downstream predictive performance is weaker, but they also produce fewer identical matches and higher DCR values. This suggests that adversarial and variational training introduce a form of implicit regularisation that reduces memorisation relative to statistical synthesis. At the same time, the two architectures exhibit distinct strengths: CTGAN preserves pairwise structure more effectively, while TVAE retains more discriminative information for downstream classification. These differences underscore that "utility" is not a single dimension but a composite of distributional, structural, and predictive properties.

The modified CTGAN and TVAE models illustrate the consequences of explicitly embedding privacy and utility considerations into the training objective. The multi-objective loss improves privacy across all metrics: identical matches decrease sharply, DCR increases, and inference risk is reduced. However, these gains come at a substantial utility cost. SDV scores decline, distributional distances increase, and downstream predictive performance drops dramatically. This confirms that privacy-aware optimisation introduces measurable distortion into the learned distribution and weakens the discriminative structure of the synthetic data. The modified models therefore represent a more privacy-preserving but less analytically useful regime.

Taken together, these findings demonstrate that synthetic data generation inherently involves balancing competing objectives. No single model dominates across all metrics, and the choice of synthesiser must be guided by the specific priorities of the application context. If analytical fidelity is the primary concern, Synthpop provides the strongest utility but requires careful privacy management. If moderate utility with moderate privacy is acceptable, the original CTGAN and TVAE models offer a balanced compromise. If privacy protection is paramount, the modified models provide stronger safeguards at the cost of reduced utility.

Finally, the results highlight the importance of transparency and context-sensitive design in synthetic data workflows. Synthetic data should not be treated as a universal privacy solution, but rather as a flexible tool whose behaviour depends on model architecture, training objectives, and hyperparameter choices. Responsible synthetic data release requires explicit evaluation of both utility and privacy, clear communication of trade-offs, and careful alignment with the ethical and analytical requirements of the intended use case.

5.1 Ethical Implications

The findings of this project carry several ethical implications for the design, evaluation, and deployment of synthetic data systems. Although synthetic data is often presented as a privacy-preserving alternative to real data, our results show that generative models do not inherently guarantee privacy, nor do they reliably preserve analytical fidelity. This challenges the common assumption that synthetic data can be used as a drop-in replacement for real data without careful consideration of its limitations.

First, the high disclosure risk observed in Synthpop highlights that classical statistical synthesisers may unintentionally reproduce real individuals when categorical variables have many levels or when leaf sizes are small. In regulated environments—such as official statistics, public-sector data release, or fairness auditing—such leakage may violate ethical norms of confidentiality and undermine public trust. Even when synthetic data appears statistically plausible, identical matches and low DCR values demonstrate that privacy cannot be assumed.

Second, the deep generative models illustrate a different ethical concern: the risk of misrepresenting minority groups. Both CTGAN and TVAE struggle with imbalanced data, leading to weaker predictive performance and distorted distributions for underrepresented subpopulations. This raises fairness concerns, particularly when synthetic data is used for downstream decision-making or model evaluation. If synthetic data amplifies existing biases or erases minority patterns, it may produce misleading insights or reinforce inequities.

Third, the modified CTGAN and TVAE models show that privacy-aware optimisation can reduce memorisation but also degrade utility. This trade-off has ethical consequences: synthetic data that is too distorted may lead to incorrect scientific conclusions, flawed policy analysis, or unreliable machine learning evaluation. Conversely, synthetic data that prioritises utility may expose individuals to inference or linkage attacks. Ethical synthetic data design therefore requires explicit communication of these trade-offs, rather than presenting synthetic data as a universal privacy solution.

Finally, the evaluation metrics themselves have ethical limitations. Metrics such as DCR, inference risk, and CM3 capture specific attack vectors but do not cover the full spectrum of privacy threats, including attribute inference, membership inference under adversarial training, or linkage attacks using external datasets. Ethical deployment of synthetic data requires acknowledging these gaps and avoiding overconfidence in metric-based guarantees.

Overall, the ethical implications of this work emphasise that synthetic data must be treated as a context-sensitive tool rather than a blanket privacy mechanism. Responsible use requires transparent reporting of utility and privacy characteristics, careful alignment with the intended application, and ongoing evaluation of risks. Synthetic data can support innovation and protect individuals, but only when its limitations are clearly understood and ethically managed.

5.2 Limitations

While this project provides a comprehensive evaluation of utility loss and privacy risks in synthetic data generation, several limitations must be acknowledged. These limitations concern the dataset, the models, the evaluation metrics, and the experimental environment, and they inform the scope within which our conclusions should be interpreted.

First, the analysis is based on a single dataset—the UCI Census-Income table—which, although widely used in synthetic data research, has specific structural properties that may not generalise to other domains. Its high-cardinality categorical variables, strong class imbalance, and substantial duplicate mass influence both utility and privacy metrics. Models that perform well on this dataset may behave differently on medical, financial, or behavioural data with different statistical characteristics.

Second, the deep generative models were trained under computational constraints imposed by the Google Colab environment. The reduced sample size, limited training epochs, and absence of hyperparameter tuning may have affected model performance. CTGAN in particular is sensitive to training instability, and more extensive optimisation could yield different results. Similarly, the patched CTGAN and TVAE implementations used for the multi-objective loss are research prototypes rather than production-grade systems, and their behaviour may differ from fully engineered variants.

Third, the privacy metrics used in this project capture only a subset of possible attack vectors. Metrics such as DCR, inference risk, and CM3 quantify memorisation and nearest-neighbour vulnerability, but they do not address attribute inference under adversarial training, membership inference using shadow models, or linkage attacks involving external datasets. Consequently, the privacy guarantees reported here should be interpreted as partial rather than comprehensive.

Fourth, the multi-objective loss function itself has limitations. The weighting coefficients were not systematically tuned, and the additional loss terms were computed without gradient tracking to avoid destabilising training. This design choice simplifies implementation but may reduce the effectiveness of the optimisation. A more principled approach—such as Pareto-optimal training, constrained optimisation, or differential privacy mechanisms—could provide stronger guarantees.

Finally, synthetic data inherently risks misrepresenting minority groups. The Census-Income dataset contains demographic imbalances that deep generative models may amplify or distort. Although this project evaluates predictive utility and structural fidelity, it does not include a fairness analysis or subgroup-specific evaluation. As a result, the impact of synthetic data on underrepresented populations remains an open question.

Overall, these limitations highlight that synthetic data generation is a context-dependent process. The results of this project provide valuable insights into utility–privacy trade-offs, but they should be applied with caution and supplemented with additional evaluation when used in high-stakes or sensitive environments.

6 Conclusion

This project set out to investigate how synthetic data generation affects analytical utility and privacy protection, and to explore whether these competing objectives can be balanced through loss-aware optimisation. Using three representative model families—Synthpop, CTGAN, and TVAE—we evaluated distributional fidelity, structural preservation, downstream predictive performance, and multiple privacy metrics on a reduced version of the UCI Census-Income dataset. The results demonstrate that synthetic data generation is fundamentally shaped by trade-offs: models that preserve utility tend to expose privacy risks, while models that protect privacy often sacrifice analytical value.

Synthpop achieved the strongest utility across all metrics, confirming the effectiveness of tree-based conditional modelling for tabular data with complex categorical structure. However, its high disclosure risk underscores that classical statistical synthesisers may reproduce real individuals when categorical variables have many levels or when leaf sizes are small. This makes Synthpop suitable for high-fidelity analytical tasks but ethically challenging in contexts requiring strong confidentiality guarantees.

The original CTGAN and TVAE models provided a more balanced profile, offering moderate utility with moderate privacy protection. Their adversarial and variational training introduced implicit regularisation that reduced memorisation relative to Synthpop, but they struggled to match Synthpop's structural fidelity and downstream predictive performance. The two models also exhibited distinct strengths: CTGAN preserved pairwise relationships more effectively, while TVAE retained more discriminative structure for classification.

The modified CTGAN and TVAE models demonstrated that privacy-aware optimisation can shift generative models toward a more protective regime. The multi-objective loss reduced identical matches, increased DCR, and lowered inference risk, indicating improved privacy behaviour. However, these gains came at a substantial utility cost: SDV scores declined, distributional distances increased, and downstream predictive performance dropped sharply. This confirms that embedding privacy and utility terms directly into the training objective introduces measurable distortion into the learned distribution.

Taken together, these findings show that no single synthesiser dominates across all metrics. Instead, synthetic data generation requires context-sensitive design: the appropriate model depends on whether utility, privacy, or a balance of both is the primary concern. High-fidelity applications may favour Synthpop; moderate-risk environments may prefer CTGAN or TVAE; and privacy-critical settings may benefit from loss-aware optimisation or stronger mechanisms such as differential privacy.

Finally, this work highlights the importance of transparency, ethical reflection, and rigorous evaluation in synthetic data workflows. Synthetic data should not be treated as a universal privacy solution, but as a flexible tool whose behaviour depends on model architecture, training objectives, and dataset characteristics. Responsible synthetic data release requires clear communication of utility–privacy trade-offs, careful alignment with the intended use case, and ongoing assessment of risks. Future work may explore differential privacy, Pareto-optimal training, fairness-aware evaluation, and multi-dataset benchmarking to further strengthen the ethical and analytical foundations of synthetic data generation.

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